

Conjugacy and $\mathcal{S}$ -graph Shifts

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1 Introduction

Let $\mathcal{A}$ be a finite set. A *shift space* is a closed set $X \subseteq \mathcal{A}^{\mathbb{Z}}$ that is invariant under the *shift map* $\sigma : \mathcal{A}^{\mathbb{Z}} \rightarrow \mathcal{A}^{\mathbb{Z}}$ defined in section two. Intuitively, the points in a shift space are an indexed list of letters, and the shift map shifts the index on each letter one to the left. There are several different classes of shift spaces. One common example are S -gap shifts. This class of shifts is defined on the alphabet $A = \{0, 1\}$, along with a set $S \subseteq \mathbb{N}$ where the number of 0's at any point in the word is some element of S . S -gap shifts are a well studied and understood class of shifts. They prove to have applications across several fields of mathematics including algebra, topology, number theory, and ergodic theory.

While $\mathcal{S}$ -gap shifts and other simpler shift spaces are well understood, the mathematician's hunger is never satisfied. Thus, there have been several attempts to generalize S -gap shifts. Matson and Sattler [3] generalize S -gap shifts to any size finite alphabet in two different ways. Namely, the *ordered* and *unordered $\mathcal{S}$ -limited shifts*. In [1] Dillon further generalizes these by defining the $\mathcal{S}$ -graph shift, which will be the class of shifts focused on in this paper. The $\mathcal{S}$ -graph shift is represented by a simple directed-graph. Each vertex corresponds to a letter in the alphabet, and each edge corresponds to an allowed transition between letters in words of the shift. In addition, each vertex is equipped with some subset of $\mathbb{N}$. Anytime a letter appears in a word, it can only appear as many times as some element of its corresponding set, similar to how $\mathcal{S}$ controls the number of zeros in the S -gap shift. $\mathcal{S}$ -graph shifts contain all $\mathcal{S}$ -limited shifts, subshifts of finite type, and several other things in between.

A topic of interest when studying shift spaces, and much of what will be focused on in this paper, involves the concept of *conjugacy*. A conjugacy is a continuous map with a continuous inverse that commutes with the shift map. Two shifts are conjugate if there exists a conjugacy between them. We can map points bijectively between spaces through a conjugacy, which allows us to establish an equivalency between the points that also preserves dynamical behavior with respect to the map. The main takeaway is that we see all important dynamical properties preserved through conjugacy. Results on conjugacy are difficult to prove, and a result on conjugacy for $\mathcal{S}$ -graph shifts is a large motivation for the work described in this paper. There are some existing results about conjugacy for some of the shifts already mentioned here. For example, in [3], Matson and Sattler prove a result on shift spaces that have a special property called a *core set*. Say we have two shift spaces X and X' where each has a core set. Matson and Sattler show that if there exists an invertible sliding block code between X and the full shift on the alphabet of X' that induces a bijection between the core sets of X and X' , then this sliding block code is a conjugacy between the shifts. This gives us some additional information and tools when determining conjugacy. That is, our conjugacy must induce a bijection between the core sets. However, this still does not give us a concrete result on conjugacy. This only gives us a better starting point when looking for a conjugacy between spaces. Additionally, not all shift spaces have a core set (such as unordered $\mathcal{S}$ -limited shifts) so we cannot make use of this result. In [4] McLane proposes an unproven generalization of the core set for $\mathcal{S}$ -graph shifts he calls the *core graph*. In McLane's paper he outlines his conjecture for the core graph, and provides some arguments for its seeming usefulness.

In this paper, we will ride the coattails of McLane and dig further into some of his proposed ideas. We start first with an investigation of two conjugate $\mathcal{S}$ -graph shifts, which illustrates how even a relatively simple example can prove to be difficult and tedious. This is an attempt to motivate the reader as to why a more concrete result about conjugacy would prove helpful.

We will follow this with a broad explanation of McLane's proposed core graph and his conjectures relating to it, with a few examples to illustrate the idea. We then deviate to several new ideas fueled by one of Dillon's results in [1], leading to several new potential avenues for investigation in the future.

2 Background and Definitions

Throughout this paper we let $\mathbb{N} = \{1, 2, 3, \dots\}$. The symbol $\mathcal{A}$ denotes a finite set of symbols called an *alphabet*. We define the *full $\mathcal{A}$ -shift* as the collection of all bi-infinite sequences of symbols from $\mathcal{A}$, denoted by $\mathcal{A}^{\mathbb{Z}} = \{(x_i)_{i \in \mathbb{Z}} : x_i \in \mathcal{A} \text{ for all } i \in \mathbb{Z}\}$. This is the general object we'll be working with.

These bi-infinite sequences are sometimes referred to as *points* in the shift. We often talk about, a *block* or *word* over A , which is a finite sequence of symbols from A . We define the *length* of a block x by the number of letters in the block, denoted $|x|$. A block of length k is a k -block, and the set of all k -blocks over A is $A^k = \{w_1 w_2 \dots w_k : w_i \in A\}$.

We define the *shift map* $\sigma : \mathcal{A}^{\mathbb{Z}} \rightarrow \mathcal{A}^{\mathbb{Z}}$ on the full shift $\mathcal{A}^{\mathbb{Z}}$ by $\sigma(x) = \dots y_{-1} y_0 y_1 \dots$ where $y_i = x_{i+1}$. Given an origin, this map essentially just shifts every letter in the word to the left. We can define a metric d on $\mathcal{A}^{\mathbb{Z}}$ by $d(\omega, \tau) = 2^{-k+1}$ where k is the greatest integer such that $\omega_{-k} \dots \omega_k = \tau_{-k} \dots \tau_k$ and $d(\omega, \tau) = 0$ if no such k exists. $(\mathcal{A}^{\mathbb{Z}}, d)$ is a compact metric space. Using the shift map, and this metric we define the notion of *shift invariance*, for some subspace X of $\mathcal{A}^{\mathbb{Z}}$. We say X is shift invariant if for all $x \in X$ there is no such $n \in \mathbb{Z}$ such that $\sigma^n(x) \in X$. Essentially we cannot shift any word in X to a word not in X . Taking our usual notion of closure (using our defined metric d) we define *subshifts* or *shift spaces* over $\mathcal{A}$ as some subspace of $\mathcal{A}^{\mathbb{Z}}$ which is both closed and shift-invariant.

There are several classes of subshifts that have been studied to various degrees. The basic way we can create a subshift is to remove points from the full shift $\mathcal{A}^{\mathbb{Z}}$ by imposing restrictions on allowable blocks in each word. Consider the following,

For each set $\mathcal{F}$ of blocks over $\mathcal{A}$, we define

$$X_{\mathcal{F}} = \{x \in \mathcal{A}^{\mathbb{Z}} : \text{no element of } \mathcal{F} \text{ appears in } x.\}$$

$X_{\mathcal{F}}$ is a subshift of $\mathcal{A}^{\mathbb{Z}}$ for any given $\mathcal{F}$. We call $\mathcal{F}$ the set of *forbidden words*. Here we're defining our subshift in terms of words that cannot occur in any point in the shift. It may be surprising to know that the subshifts defined by forbidden words are exactly the subshifts of $\mathcal{A}^{\mathbb{Z}}$. So every subshift can be represented by some set of forbidden words. This idea of restricting words is the baseline for how we build subshifts. A basic example of a more broad class of shifts is a *subshift of finite type*, which is a subshift $X = X_{\mathcal{F}}$, such that $\mathcal{F}$ can be written as a finite set. Many subshifts have $\mathcal{F}$ such that $\mathcal{F}$ cannot be represented as a finite set.

We now define several shifts based on a similar idea to this sets of forbidden words, but with a bit of a different flavor. We impose restrictions on how many of a specified letter may appear in a row, which implicitly forbids sets of words. Consider the following,

Definition 2.1. For each $S \subseteq \mathbb{N}_0$, the S -gap shift $X(S)$ is the subshift of $\{0, 1\}^{\mathbb{Z}}$ obtained by taking the closure of the set

$$\{\dots 10^{s_{-1}} 10^{s_0} 10^{s_1} 1 \dots : s_i \in S \text{ for each } i \in \mathbb{Z}\}$$

Example 2.2. (Even Shift) Taking $S = 2\mathbb{N}_0$, we call $X(S)$ the even shift. Words in this shift look then something like $10^{2k}1$ and we will never see something like $10^{2k+1}1$

This example displays how these differ from subshifts of finite type. If we consider the even shift in terms of its forbidden word list, we see we will need an infinite set, all possible $10^{2k+1}1$ words to cover all forbidden strings on the shift. Thus we will have an infinite set of forbidden words.

This idea of defining a set dictating how much of each letter in our alphabet can appear in tandem across any word in a shift will stay with us for the rest of this paper. Here we work only with a fixed alphabet $0, 1$ and only define a set on 0 . We can extend this idea to an arbitrary alphabet, with sets on each letter that dictate the allowed frequency of each letter when it appears.

Definition 2.3. Let $\mathcal{A} = \{1, \dots, n\}$, and for $n \geq 2$, $\mathcal{S} = (S_1, \dots, S_n)$ be an n -tuple of nonempty subsets of $\mathbb{N}$. We set $G_{\mathcal{S}} = \{1^{m_1} 2^{m_2} \dots n^{m_n} : m_i \in S_i \text{ for each } i \in \mathcal{A}\}$, called the *core set*. The *ordered $\mathcal{S}$ -limited shift*, denoted $\vec{X}(\mathcal{S})$, is the subshift of $\mathcal{A}^{\mathbb{Z}}$ obtained by taking the closure of the set

$$\{\dots x_{-1} x_0 x_1 \dots : x_i \in G_{\mathcal{S}} \text{ for each } i \in \mathbb{Z}\}$$

This notion of “*ordered*” comes from the structure of our defined core set. Namely, note that each x_i is an element of the core set, and by the way its defined will be a string that is in some sense ordered based on the order of the natural numbers. We define the *unordered $\mathcal{S}$ -limited shift*, denoted $X(\mathcal{S})$, by dropping this ordering. This is the subshift of $A^{\mathbb{Z}}$ obtained by taking the closure of

$$\{\cdots a_{-1}^{m_{-1}} a_0^{m_0} a_1^{m_1} \cdots : a_i \in \mathcal{A}, m_i \in S_{a_i}, \text{ and } a_{i+1} \neq a_i \text{ for each } i \in \mathbb{Z}\}.$$

It is worth noting that these two shifts coincide when $A = \{0, 1\}$

Example (Continuation of example 2.2).

Consider again the even shift $X(S)$ with $S = 2\mathbb{N}_0$. Note this is actually an ordered $\mathcal{S}$ -limited shift on the alphabet $A = \{0, 1\}$ and $S = \{2\mathbb{N}, \mathbb{N}\}$. We must pick our sets this way, due to the fact that the $\mathcal{S}$ -gap as it is defined uses $\mathbb{N}_0$. This allows for infinite strings of ones, which we must consider when writing this as an ordered $\mathcal{S}$ -limited shift. With this information we can also explicitly write out our core set:

$$G_s = \{0^{2k}1^n : k \in \mathbb{N}_0, n \in \mathbb{N}\}$$

Note that we can build any word in the shift by indiscriminately concatenating words from this set G_s , which is how we often think about the core set.

Both ordered and unordered $\mathcal{S}$ -limited shifts are two specializations of the primary type of subshift we’re considering in this paper, the $\mathcal{S}$ -graph shift. To formally define this shift we need to first cover a bit of basic graph theory.

Moving forward, every graph in this paper will be finite, directed, and simple (no loops and no multi edges). We denote the edge and vertex sets of a graph G as $E(G)$ and $V(G)$ respectively. We call a vertex of G *essential* if its in-degree and out-degree are both at least one. The graph G is *essential* if all its vertices are. We can obtain an underlying essential graph of G by iteratively removing all non-essential vertices. A *walk* on G is a finite sequence of vertices $(v_1, \dots, v_n)$ such that $(v_i, v_{i+1}) \in E(G)$ for each $i \in \{1, 2, \dots, n-1\}$. We call G *irreducible* if there is a walk from u to v for every $u, v \in V(G)$. This essentially means that when starting from any vertex, there is some way to traverse the edges to get to any other vertex. A walk is called *closed* if $n \geq 2$ and $(v_n, v_1) \in E(G)$. A closed walk is a *cycle* if all vertices in the walk are distinct. With this background, we can now define the $\mathcal{S}$ -graph shift

Definition 2.4. A *set-equipped graph* is a pair $(G, \mathcal{S})$ where G is a directed graph and $\mathcal{S} = (S_v)_{v \in V(G)}$ is an indexed collection of nonempty subsets of $\mathbb{N}$. If $\mathcal{G} = (G, \mathcal{S})$ is a set-equipped graph, the $\mathcal{S}$ -graph shift $X(\mathcal{G}) = X(G, \mathcal{S})$ is the closure (in the full shift $V(G)^{\mathbb{Z}}$) of the set

$$\{\cdots u_{-1}^{n_{-1}} u_0^{n_0} u_1^{n_1} \cdots : u_i \in V(G), n_i \in S_{u_i}, \text{ and } (u_i, u_{i+1}) \in E(G) \text{ for each } i \in \mathbb{Z}\}.$$

Example 2.5. Let $\mathcal{A} = \{1, 2, 3\}$, $\mathcal{S} = \{S_1, S_2, S_3\}$ with $S_1 = \mathbb{N}$, $S_2 = \{1, 2\}$, and $S_3 = \{2\}$, and G be the graph given in Figure 1.

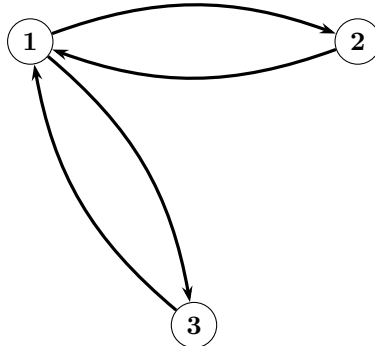


Figure 1: Graph G for Example 2.6

The $\mathcal{S}$ -graph shift is $X(G, \mathcal{S})$ and words in the shift are bi-infinite sequences determined by this set equipped graph. The directions of the edges dictate allowed transitions between letters within words of

the shift. To illustrate this consider the following two examples. The word 111112233 is an allowable word in this shift. Based on our sets, we can have seven 1's, two 2's, and two 3's. Additionally, we can go from 1 to 2 and 2 to 3 based on our directed edges. The word 222333 however is not allowed. Note this violates both our sets, and our edges. We cannot have three 2's or three 3's. Additionally, we cannot go from 2 to 3 based on our directed edges. This interplay between the edges and sets is exactly how words are determined in an $\mathcal{S}$ -graph shift.

The $\mathcal{S}$ -graph shift turns out to be a generalization for several other types of subshifts. It is relatively trivial that every SFT is conjugate to $\mathcal{S}$ -graph shift. Moreover, if the graph G of $X(G, S)$ is a directed cycle, the shift is then the ordered $\mathcal{S}$ -limited shift. If G is a complete graph, that is every vertex has an edge to every other vertex, then $X(G, S)$ is the unordered $\mathcal{S}$ -limited shift. These are just the extreme cases, though, as there are many interesting $\mathcal{S}$ -graph shifts that lie in between these cases.

The $\mathcal{S}$ -graph shift is the shift we'll be looking at primarily through the course of this paper. First however, we need to take a step back and talk about some more general properties of shifts, building up to a specific result on $\mathcal{S}$ -graph shifts.

The primary property we are interested in in relation to $\mathcal{S}$ -graph shifts is the subject of the following definition,

Definition 2.6. Let X and Y be two subshifts. We say that X and Y are conjugate (notated as $X \cong Y$) if there exists a homeomorphism $\varphi : X \rightarrow Y$ such that $\varphi \circ \sigma = \sigma \circ \varphi$ where σ denotes the shift map.

This definition is a bit broad, but a few helpful results tells us that this homeomorphism (called a *conjugacy*) must actually be a special type of function called a *sliding block code*,

Definition 2.7. Let X be a shift space, $m, n \in \mathbb{N}_0$, and $\Phi : \mathcal{B}_{m+n+1}(X) \rightarrow \mathcal{A}$. The *sliding block code* with memory m and anticipation n , also called an $(m+n+1)$ -block code, is the map $\phi : X \rightarrow Y$ defined by

$$\phi(x)_i = \Phi(x_{i-m} \cdots x_{i+n}).$$

In particular, in [2] Lind and Marcus show *conjugacy* between two shifts must be an invertible sliding block code. This gives us a bit more specificity on the nature of our conjugacies, and reduces the question of conjugacy to the existence of an invertible sliding block code. However, this is still not an easy task to tackle.

The $\mathcal{S}$ -gap shift is one of the only shifts where we have a strong result about conjugacy. That is, two $\mathcal{S}$ -gap shifts $X(S)$ and $X(S')$ are conjugate if and only if $s = \{0, n\}$ and $S' = \{n, n+1, n+2, \dots\}$ for $n \in \mathbb{N}$. This is a very powerful result about conjugacy, and something like this is the overarching goal when approaching conjugacy with different classes of shifts. But something this nice is very difficult to achieve. Even a result such as the one found in [3] by Matson and Sattler stated in the introduction, which only gives us an additional tool to consider when determining conjugacy, can be evasive.

A large part of the study of conjugacy includes the investigation of special properties of a shift that are preserved through conjugacy. We call these properties *invariants*. There are several of these invariants that exist. A point $x \in A^{\mathbb{Z}}$ is called *periodic* with period n if $\sigma^n(x) = x$. $p_n(X)$ defined as the number of points with period n in the shift X is an invariant. Similarly, we can define points with *least period* n , and see that the number of these points is also an invariant. However, the invariant worked with most over the course of this project is *entropy*.

Definition 2.8. The (*topological*) *entropy* of a shift space X is

$$h(X) = \lim_{n \rightarrow \infty} \frac{1}{n} |\text{Log}(B_n(X))|$$

The entropy is a measure of the size of a shift space's language, which in some sense provides an idea of the complexity of the space. While you can calculate the entropy using this definition, there are several formulas that provide a way to calculate the entropy for specific types of subshifts. Dillon [1] proved the following formula for the entropy of an $\mathcal{S}$ -graph shift.

Letting $\mathcal{C}_G$ denote the collection of all nonempty sets of vertex disjoint cycles in a graph G . That is, each element of $\mathcal{C}_G$ is itself a set of cycles in G which do not share a vertex. This will include all the singleton sets for each cycle in the graph. For $C \in \mathcal{C}_G$, we let $|C|$ denote the cardinality of C . Finally, we let $H_c(X)$ denote the generating function associated with the cycle c . With this we state,

Theorem 2.9. *Let (G, S) be a set-equipped graph. If λ is the least positive solution to*

$$\sum_{C \in \mathcal{C}_G} (-1)^{|C|+1} \prod_{c \in C} H_c(X) = 1$$

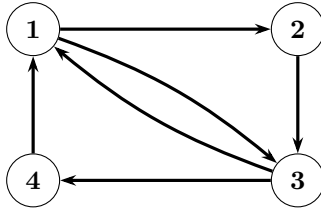
then the entropy of $X(G, S)$ is $\log \lambda^{-1}$

This theorem will be the basis for much of what we do moving forward in this paper.

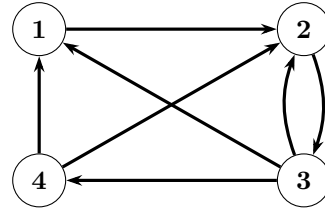
3 A motivating example

We mentioned how our overall goal was to get a better understanding of conjugacy between $\mathcal{S}$ -graph shifts, and best case scenario find some sort of necessary and sufficient condition for conjugacy. In this section we'll provide some motivation for this goal. We show a conjugacy simply using the tools we've built up in the background, demonstrating that it is not a straightforward process.

Example 3.1. Consider the following two $\mathcal{S}$ -graph shifts $X_1 = X(G_1, S^1)$ and $X_2 = X(G_2, S^2)$ on the alphabet $A = \{1, 2, 3, 4\}$ with $S^1 = \{\{1, 3\}, \{\mathbb{N}\}, \{3\}, \{1\}\}$, $S^2 = \{\{\mathbb{N}\}, \{1, 3\}, \{3\}, \{1\}\}$



(a) Set equipped graph G_1



(b) Set equipped graph G_2

In the world of $\mathcal{S}$ -graph shifts, this is actually a relatively simple example. Our sets are simply permutations of each other, and each graph contains a 4-cycle, a 2-cycle, and two 3-cycles. However, the rest of this example will show that determining conjugacy even in a simple case may not be immediate. Our goal will be to build an invertible sliding block code between these graphs. In a real scenario, we would not jump straight to this step. Rather, we would need to build some intuition about whether or not these may be conjugate. This step usually consists of checking some intuitive properties, as well as computing invariants. Typically we will compute several invariants to see if they line up before attempting to build a conjugacy. Here, we will just check the entropy. This gives just a piece of what this part of the process may look like, as well as providing an example of using Dillon's entropy formula introduced earlier.

To compute the entropy with our formula, we first must determine $\mathcal{C}_{G_1}$ and $\mathcal{C}_{G_2}$. Following as stated in Dillon's formula, we have:

$$\mathcal{C}_{G_1} = \{\{1234\}, \{123\}, \{134\}, \{13\}\}$$

$$\mathcal{C}_{G_2} = \{\{1234\}, \{123\}, \{234\}, \{23\}\}$$

Note for a shift $X(G, S)$ we define generating functions for on each vertex $v \in V(G)$ as $H_v(t) = \sum_{s \in S_v} t^s$. When considering a cycle $c = v_1 v_2 \dots v_n$ we then can define the generating function of cycle as $H_c(t) = H_{v_1}(t) * H_{v_2}(t) * \dots * H_{v_n}(t)$. With this we can move forward.

For X_1 we have generating functions

$$\begin{aligned} H_1(X) &= t + t^3 & H_2(X) &= \frac{t}{1-t} \\ H_3(X) &= t^3 & H_4(X) &= t \end{aligned}$$

Similarly, for X_2 we have,

$$\begin{aligned} H_1(X) &= \frac{t}{1-t} & H_2(X) &= t + t^3 \\ H_3(X) &= t^3 & H_4(X) &= t \end{aligned}$$

Noting now that our disjoint cycle sets all have cardinality one, and thus all our terms based on our formula will be positive, we can move forward with our calculation. For X_1 our equation becomes,

$$\frac{t^6 + t^8}{1-t} + \frac{t^5 + t^7}{1-t} + t^5 + t^7 + t^4 + t^6 = 1$$

for X_2 we have,

$$\frac{t^6 + t^8}{1-t} + \frac{t^5 + t^7}{1-t} + t^5 + t^7 + t^4 + t^6 = 1$$

Note that in both cases we get the exact same equation. Thus we can stop here as both will clearly have the same least positive solution, resulting in the same entropy. This example works out rather nicely as we end up with the exact same equation. One can imagine however, that this may not be such a simple task as we get more complicated shifts, with more complex generating functions. Additionally, it is not always the case that the equation generated for calculating entropy in Theorem 2.10 is the same between two conjugate shifts. Rather, they may just have the same least positive solution. So this must always be explicitly checked. With this in mind one could imagine how even this process of checking invariants may be frustrating.

Now let us build the sliding block code. These graphs have a nice property that makes our job slightly easier. Namely, each shift can be 'built' out of a single vertex. When we have this property we can find a distinct set of building blocks for each shift, which will form essentially a core set. We can leverage this and build our sliding block code to bijectively map between each core set. Written explicitly our core sets are,

$$G_{S^1} = \{12^n 3334, 1112^n 3334, 12^n 333, 1112^n 333 : n \in \mathbb{N}\} \cup \{1333, 111333, 13334, 1113334\}$$

$$G_{S^2} = \{1^n 23334, 1^n 2223334, 1^n 2333, 1^n 222333 : n \in \mathbb{N}\} \cup \{2333, 222333, 23334, 2223334\}$$

Note that we use powers of n to indicate that we can have any number $n \in \mathbb{N}$ of that letter (in this case 1 or 2) based on the set definitions. Due to this fact we are actually working on infinite sets. Note also that each of the blocks in each set contain the 'single' vertex briefly mentioned earlier (in which we can build the shift) that appears in every cycle, (namely 1 and 2 respectively).

All that is really happening in this example is 1 and 2 are swapping roles. The sets on 3 and 4 stay fixed, and the strings in each core set look very similar besides the swapping of 1 and 2's roles in dictating the length of each block. However, the sliding block code will still turn out to be rather complex. Looking at each core set we can see relatively easy the candidates for which element should map to which. We are looking to map cycles to cycles, and map words of the same length to each other. Using this, we can imagine we're mapping each 4, 3, and 2 cycle to each other. Additionally, since one of our sets is not a singleton, we have several words corresponding to any cycle that containing this letter. So for example our 2-cycle 13 on G_1 corresponds to two words. We will be mapping this to the 2-cycle 23 on G_2 , which also corresponds to two words. Each cycle contains information about two words. We will map a cycle to a cycle, and then within each cycle map the words of equal length to each other. Following this idea, we need a sliding block code $\Phi : \mathcal{B}_{m+n+1}(X_2) \rightarrow \mathcal{A}$ that induces a bijection $\phi : X_2 \rightarrow X_1$

$$\begin{aligned}
\phi(12^n 3334) &= 1^n 23334 \\
\phi(1112^n 3334) &= 1^n 2223334 \\
\phi(1333) &= 2333 \\
\phi(111333) &= 222333 \\
\phi(12^n 333) &= 1^n 2333 \\
\phi(1112^n 333) &= 1^n 222333 \\
\phi(13334) &= 23334 \\
\phi(1113334) &= 2223334
\end{aligned}$$

We will now construct our sliding block code. When doing this, we will consider only strings with a single 1 (and correspondingly a single 2). We can relatively easily adjust the code to account for when we have multiple ones (i.e. any time we have an extra one, we will output an extra two). Here we take the case of just one of these letters appearing to find our initial rule. Now, consider the sliding block code Φ with memory $m = 3$ and anticipation $n = 1$ defined by

$$\begin{aligned}
\Phi(w_1 w_2 w_3 w_4 w_5) &= w_4 \text{ with some special cases,} \\
\Phi(33122) &= 1 \quad \Phi(42223) = 1 \quad \Phi(32223) = 1 \\
\Phi(31222) &= 1 \quad \Phi(34122) = 1 \quad \Phi(33423) = 1 \\
\Phi(33323) &= 1 \quad \Phi(33422) = 1 \quad \Phi(33322) = 1 \\
\Phi(41222) &= 1 \quad \Phi(34222) = 1 \quad \Phi(33222) = 1
\end{aligned}$$

Note our special cases all involve swapping mapping a 2 to a 1. There is a lot of intricacy happening here to ensure this mapping works based on the structure of the shifts. This can be part of the difficulty in finding a conjugacy.

We will not explicitly check that our sliding block code works for each element of the core set, but we will do one for good measure. Consider the word 13334. We will start mapping from 1, but in order to incorporate our memory and anticipation we will imagine that we are seeing this string repeated as 133341333413334,

Starting at our middle 1 applying our sliding block code we have,

$$\begin{aligned}
33413 &\rightarrow 2 \quad (1333 \underline{33413} 3413334) \\
34133 &\rightarrow 3 \quad (13334 \underline{34133} 413334) \\
41333 &\rightarrow 3 \quad (133341 \underline{41333} 13334) \\
13334 &\rightarrow 3 \quad (1333413 \underline{13334} 13334) \\
33341 &\rightarrow 4 \quad (13334133 \underline{33341} 3334)
\end{aligned}$$

Note that this was our desired mapping as stated in terms of ϕ above. Thus it seems our code works properly, at least in this case.

Hopefully this example provides some motivation on determining an easier way to approach conjugacy between $\mathcal{S}$ -graph shifts. There were a lot of moving parts here, in particular for the construction of the sliding block code. It ended up being rather complex in order to catch the different transitions possible between the cycles that contained the ‘infinite’ letter and those that did not. But even with that intricacy, this is still a relatively simple example. We were able to obtain a core set, which allowed us to hone our process of developing a sliding block code by ensuring it was a bijection between these sets.

In general, there is no sure way to determine whether an $\mathcal{S}$ -graph shift has a core set, and indeed it is a known fact that there will not always be a core set (for example when the shift represents an unordered $\mathcal{S}$ -limited shift). So this simpler approach used in this example won't always be applicable. This process of checking invariants, and manually constructing a conjugacy leaves the interested begging for a more convenient way to approach on conjugacy between $\mathcal{S}$ -graph shifts. We know that cycles must map to cycles, with the associated words on each cycle matching lengths as well. With this idea in mind, we move forward in this paper describing some attempts at approaching, and obtaining some ideas to contribute towards this goal.

4 The Core Graph

In [4] McLane proposes a construct he calls the *core graph*. The goal of the core graph is to generalize the idea of a *core set* to all $\mathcal{S}$ -graph shifts. Recall that the core set is a set of blocks in a shift that can be concatenated together indiscriminately to create any word in the shift. $\mathcal{S}$ -graph shifts can be complex, with several allowed blocks and transitions. This can make some notion of a core set where we can indiscriminately concatenate blocks impossible. However, we may be able to find blocks that can be effectively used to build the shift given that we are careful about how we concatenate them. This is the information used to construct the core graph.

McLane's proposed structure is a directed graph, where each vertex represents a cycle on the graph (which is indeed an allowable word of the shift). The idea here is that the cycles are themselves the building blocks we need. However, it is not true that we simply throw all cycles on the core graph and call it a day. Rather, there may be some cycles that are actually compositions of other cycles, and in turn may be omitted. With each vertex representing an allowable word of the shift, the directed edges between them then describe how we can concatenate these blocks. So rather than indiscriminate concatenation, we must ensure to concatenate based on these edges (similar to how edges on the $\mathcal{S}$ -graph shift dictate concatenation of letters). In the case that a shift has a core set, this core graph would then be a complete digraph, where each vertex corresponds to some element in the core set. The completeness of the graph then represents the indiscriminate concatenation described with core sets.

Thinking back to our previous example, the core graphs for each of these shifts would be complete with 8 vertices (note, some of them actually represent infinite blocks. McLane kept the exponential notation in each vertex to cover this), each representing one element of the core set. McLane's conjectured that two $\mathcal{S}$ -graph shifts are conjugate if and only if their core graphs are graph isomorphic, with identified vertices representing words of the same length. This would clearly be the case with our previous example, as each element in one core set maps to an element in the other with the same length. This though is a bold claim to make, especially considering the lack of rigor in the proposed definition (or really lackthereof) of the core graph. The process McLane used for finding the core graph is not clearly defined, and does not yield unique results (due to an issue about choice of orientation to be touched on later).

An additional difficulty of the core graph as proposed is the way it captures information about the sets. Recall that each vertex in the core graph corresponds to some word associated with a cycle. So if some letter on the original graph G has more than one element in its associated set, then there will be several vertices on the core graph for each single cycle containing that vertex. Namely one for each entry in the set. Note that if several letters are associated with sets containing several elements, things start to blow up. The number of vertices corresponding to a single cycle will then be the product of the cardinalities of the associated sets for each letter in the cycle. This can quickly yield unwieldy core graphs, that may be even harder to work with than the original shift. Due to this, and the general lack of an algorithmic definition for determining the core graph, we have not been able to test its usefulness for sufficiently complicated examples.

With that being said, we still continued forward working with the core graph in mind. It seemed to have potential as a useful tool, at least for our simpler examples. We worked through some examples to see if they support McLane's conjecture about conjugacy and the core graph. Consider the following graphs in figure 3 and their corresponding core graphs in figure 4 (which we provide without explanation

of McLane's process). We must note that we're omitting the discussion of the sets here. The core graphs as they are given here represent all sets on each vertex in the shifts from figure 3 being the singleton 1. Looking at the graphs in this way still provide some useful insight into the ideas of the core graph.

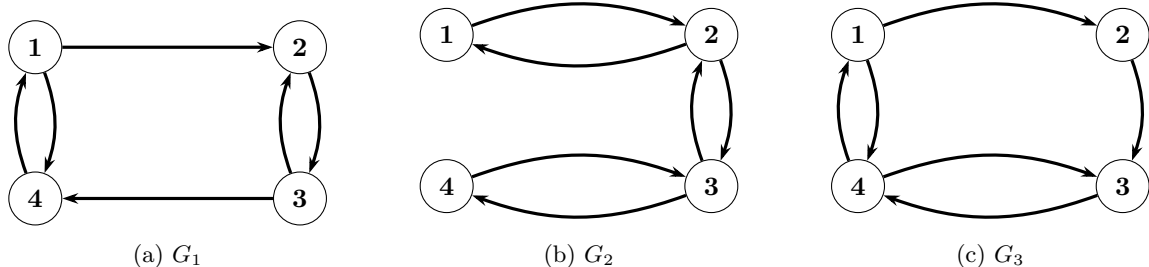


Figure 3: Three directed graphs

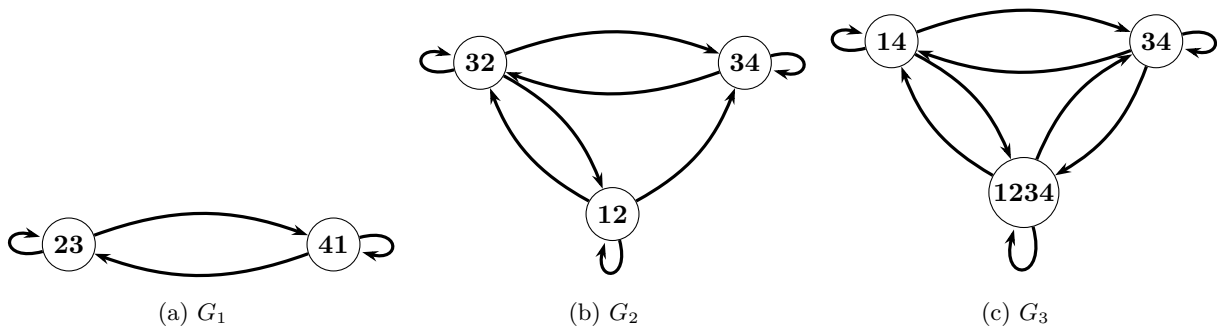


Figure 4: Associated core graphs

Note then by McLane's conjecture, none of these shifts would be conjugate, as none of these core graphs are graph isomorphic. We ought to be careful again in saying this, as we actually need to assign sets to each of these graphs to make them $\mathcal{S}$ -graph shifts. If we kept singleton sets on all the vertices for the graphs in figure three, then the core graphs would have the same structure as those provided in figure 4. Thus the claim would still be no existence of a conjugacy. We did not verify rigorously this lack of conjugacy between these shifts, but several checks seemed to support this conclusion.

Assuming McLane's conjecture, these core graphs do suggest it may be possible to find sets that make G_1 and G_3 conjugate. That is, consider what may happen to the core graph of G_1 if one of the vertices had an associated set with cardinality 2. Then the cycles containing that vertex would correspond to two words, and thus would add an additional vertex on the core graph. If chosen properly, it may produce an isomorphic core graph to that of G_3 , with identified vertices corresponding to words of the same length. We did indeed find a conjugacy in this way. Note however that we can *not* do something like this for G_2 , as its core graph has a missing edge. Thus there is no way to pick sets to produce an isomorphic core graph to that of G_1 or G_2 , as we will always have this issue of an edge missing.

Here we can kind of see the idea captured by the core graphs. The graphs on the left and right are both complete, so these presumably represent the notion of a core set. So we can indiscriminately concatenate the words represented by these vertices. Consider though the missing edge on the core set for G_2 . This represents a disallowed transition. Namely, note that in G_2 you can transition from 1 to 2, then from 2 to 3 and 3 to 4. But in order to transition back to 1 you must first go through 3 again. That is the information captured by this missing edge. Note however the portrayal of this information is reliant on the orientation imposed by the vertices in the core graph. Namely, when you leave a vertex on the core graph you're transitioning from the *last* letter in its associated word. Similarly, when entering a vertex you're transitioning to the *first* letter of its associated word.

This idea of orientation is crucial to the proposed structure. However, it also is a source of difficulty

when looking to add any rigor to the conjecture. Orientation in this way is difficult to generalize, and define. Additionally, this notion of orientation is what breaks the uniqueness of the core graph, as there could be several different representations of the core graph (based on McLane's process and proposed definition) with each being dependent on what orientation you choose when determining the core graph. These choices of orientation were much of what made the process of obtaining the core graph very difficult.

We have described the core graph, and given some examples under the lens of McLane's strong conjecture on conjugacy. However, we have also described several issues with the core graph as it is currently proposed. These issues suggest that this conjecture is bold, and we cannot really ground it with the current lack of rigor. However, the core graph still has potential to be a useful tool. We may be able to use this as some sort of conjugacy invariant, or additional check when considering conjugacy. It could also tell us which cycles are important to consider when attempting to build a conjugacy. Additionally, something as significant as a missing edge may be a more clear marker for a lack of conjugacy, as this represents a fundamentally different dynamical behavior related to the concatenation of cycles in the shift. But if we are to have the core graph be useful at all, we really need to have a more reliable way to compute it.

5 A shift to disjoint cycles

Recall Theorem 2.10 on the entropy of an $\mathcal{S}$ -graph shift:

Theorem 5.1. *Let (G, S) be a set-equipped graph. If λ is the least positive solution to*

$$\sum_{C \in \mathcal{C}_G} (-1)^{|C|+1} \prod_{c \in C} H_c(X) = 1$$

then the entropy of $X(G, S)$ is $\log \lambda^{-1}$

Consider applying this theorem to calculate the entropy of each the shifts shown in the previous section. Let us first write our disjoint cycle sets for each of these graphs.

$$\begin{aligned}\mathcal{C}_{G_1} &= \{\{1234\}, \{23\}, \{14\}, \{23, 14\}\} \\ \mathcal{C}_{G_2} &= \{\{12\}, \{23\}, \{34\}, \{12, 34\}\} \\ \mathcal{C}_{G_3} &= \{\{1234\}, \{14\}, \{34\}\}\end{aligned}$$

Note for G_3 when we begin to build the equation described in the theorem we will get a simple sum of polynomials and/or rational functions. For G_1 and G_2 however things will be more interesting. Note we will begin getting negative terms due to some of our disjoint cycle sets having cardinality 2. Note for G_1 , our first disjoint cycle set contains only the cycle 1234, and so the first term of our equation will be

$$(-1)^{1+1} H_{1234}(X) = H_1(X) * H_2(X) * H_3(X) * H_4(X)$$

Note similarly the term that will come from our last disjoint cycle set, $\{23, 34\}$ will be

$$(-1)^{2+1} (H_{23}(X) * H_{14}(X)) = (-1)^{2+1} (H_1(X) * H_2(X) * H_3(X) * H_4(X))$$

So these terms will actually cancel each other out in the equation for the entropy. In fact, any time we have something similar to this (i.e when we have an odd cardinality cycle set, and an even cardinality cycle set), where all the vertices present are the same, then the corresponding terms in the equation will cancel. Note that for G_2 we do not have this behavior, so the generating function corresponding to the set $\{12, 34\}$ will appear in our equation, and be negative based on the cardinality of this set and Dillon's formula.

To be concise, after cancellation the cycle sets that contribute to the entropy calculation for each respective shift are

$$\{23\}, \{14\} \text{ for } G_1, \{12\}, \{23\}, \{34\}, \{12, 34\} \text{ for } G_2, \text{ and } \{1234\}, \{14\}, \{34\} \text{ for } G_3$$

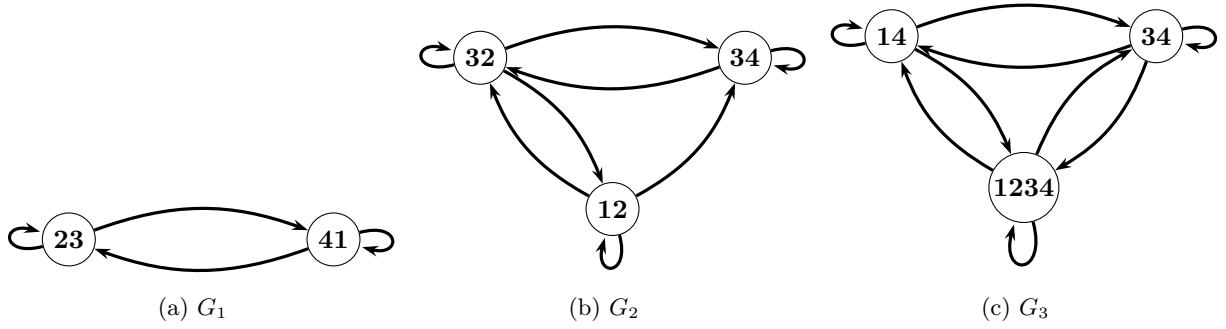


Figure 5: Associated core graphs for figure 3

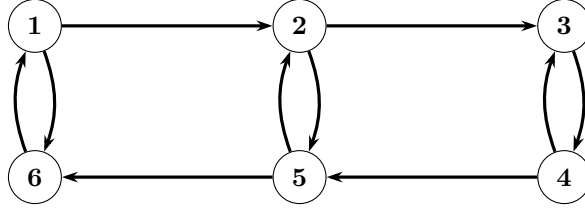
Noting again that $\{12, 34\}$ will correspond to a negative term in our equation. The attentive reader at this point may have noticed something interesting. Reconsider the core graphs provided in the previous section. Note that the vertices on each of these core graphs correspond to the non-canceled singleton cycle sets that appear in our entropy calculation. From this it appears that our non-canceled disjoint cycle sets may be telling us something about the core graph. In particular the behavior on G_1 and G_2 seemed very exciting. It seems as though for G_1 these sets tell us which cycle to omit from our core graph. Even more interestingly, note the non-canceled cycle set $\{12, 34\}$ for G_2 . On the core graph, there is an edge missing between the vertices corresponding to the two cycles 12 and 34. This suggests a possibility that non-canceled non-singleton cycle sets in our entropy formula correspond to missing edges in the core graph (and thus ways we cannot concatenate cycles).

Based on this, it appears at first glance that we may have stumbled on an easier way to compute McLane’s conjectured core graph. That is, rather than engaging in the vague and tedious process McLane had developed prior, we can instead look at the disjoint cycle sets of the shift. We consider which sets ‘cancel’ based on the structure of the described equation in the theorem. The non-canceled singletons then dictate the vertices of the core graph. And any non-canceled sets of several cycles suggested some sort of missing edge between those two associated vertices in the core graph. The trouble now was attempting to determine how exactly to remove this edge based only on this set. In the case of G_2 we must examine how the graph actually looks, and allowed transitions to determine which edge to omit. This still would depend on chosen orientation though, which may pose some difficulty. But overall, this showed promise as an idea to push the core graph further as a potential tool.

We worked through several examples with this idea in mind. It seemed to line up for some examples. However for others, there were too many intricacies with finding the core graph to verify this conjecture. And for some examples, missing edges of the core graph did not line up with non-canceled sets. Due to the lack of rigor in the process of finding the core graph, we have no way to know if we may have made a mistake in any of these examples either. However, this is not to suggest that this idea is a lost cause. If more work can be done refining the core graph, it’s possible that the idea presented here may give us information for certain classes of $\mathcal{S}$ -graph shifts, or possibly even a way to classify them. Either way, this is an open idea worth further investigation.

6 Shifting away from the core graph?

Whether the idea discussed in the previous section leads to further discovery or not, looking in to the disjoint cycle sets seemed to open up several other potential ideas for approaching conjugacy of $\mathcal{S}$ -graph shifts. These disjoint cycle sets seem to hold a lot of information about the shift. At the very least, they are closely related to the entropy as suggested by their usage in calculating it. This alone suggests they may be a powerful tool in further understanding properties of $\mathcal{S}$ -graph shifts. We’ll now look at one particular example we spent a lot of time examining through the lens of disjoint cycles/cycle sets, and pose several questions which warrant further investigation. Consider the following graph.



The disjoint cycle sets on this graph are as follows,

$$\mathcal{C}_G = \{\{123456\}, \{1256\}, \{2345\}, \{16\}, \{25\}, \{34\}, \{1256, 34\}, \{2345, 16\}, \{16, 25\}, \{16, 34\}, \{25, 34\}, \{16, 25, 34\}\}$$

Note this is the first time we're encountering a disjoint cycle set of size 3. However, cancellation works out similarly in the equation, keeping in mind how the generating functions are calculated, and which terms are positive or negative based on set cardinality. Using this, we see the following cycles cancel each other in our equation for the entropy,

$$\begin{aligned} &\{123456\} \text{ and } \{2345, 16\} \\ &\{1256\} \text{ and } \{16, 25\} \\ &\{2345\} \text{ and } \{25, 34\} \\ &\{16, 25, 34\} \text{ and } \{1256, 34\} \end{aligned}$$

This leaves us then with the sets

$$\{16\}, \{25\}, \{34\}, \{16, 34\}$$

Question 6.1. What do these disjoint cycle sets, and corresponding cancellation tell us about composition of cycles? Furthermore, can we use this to refine which cycles to consider when attempting to find conjugacy?

We ask ourselves what exactly these disjoint cycle sets may be telling us about the shift. When discussing these ideas in the context of the core graph conjecture, it seems that our non-canceled cycles give us the “core cycles” which can be used as building blocks for our shift. When looking closer at the cycles that are canceled, it seems to be related to some sense of cycle nesting or decomposition. That is, note how these smaller cycles that were not canceled are embedded within our larger 4 and 6 cycles, which are canceled. This suggests that one may want to focus only on the most ‘internested’ cycles in the shift, as they seem to be the ones directly related to the entropy (and thus hopefully the core dynamical properties of the shift). This allows us to potentially hone our view of what cycles to consider/map between when investigating conjugacy between two $\mathcal{S}$ -graph shifts.

Turning our attention to the non-canceled set of multiple cycles, it appears that this may give some information about ‘extra disconnectedness or disjointness.’ That is, the cycles $\{16\}, \{25\}, \{34\}$ are all disjoint, but $\{16, 34\}$ are in a sense more disjoint. One way to think about this is to look at the graph and consider what a word may actually look like. For example, whenever we have a 2 or a 5 in a word, it may be followed or preceded directly by the letters 1, 6, 3, or 4 (taking care to follow allowed transitions). However, any time there is a 1 or a 6, before we can see a 3 or a 4, we must see a 2 or a 5. In other words, the letters 1/6 will never appear next to 3/4. In this sense the words that these two cycles represent are words that will never be directly concatenated together. It is possible that these disjoint cycle sets with several cycles tell us information about words like this. This could be helpful in determining which cycles map to which in two conjectured conjugate shifts. This also perhaps could be some sort of invariant, where conjugate shifts must have matching non-canceled disjoint cycle sets.

Question 6.2. Do non-canceled non-singleton disjoint cycle sets tell us how we can swap sets between two $\mathcal{S}$ -graph shifts on the same graph G while maintaining conjugacy?

Note for a large majority of this paper we have been working with $\mathcal{S}$ -graph shifts arbitrarily given only with their graph G , but not their associated sets. Thinking of the shifts like this can often simplify

processes for attempting to work through some general properties. However, these sets are something you cannot ignore when fully considering conjugacy. A question we might consider is the following. Given a graph G , in what ways can we find two different sets S^1 and S^2 such that $X(G, S^1) \cong (G, S^2)$. Despite being on the same graph, finding two sets that make these shifts conjugate is not necessarily trivial.

Often when considering a similar problem in the context of say $\mathcal{S}$ -limited shifts, we consider sets that are either permutations of each other, or one is determined from the other by a sort of 'bumping' or shifting elements between the sets in the set. This is how we will approach this problem. To illustrate how this idea works, consider the following simple example,

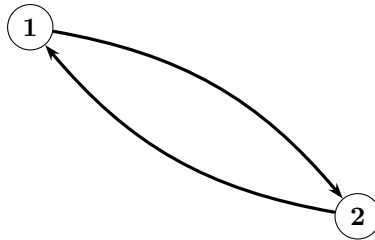
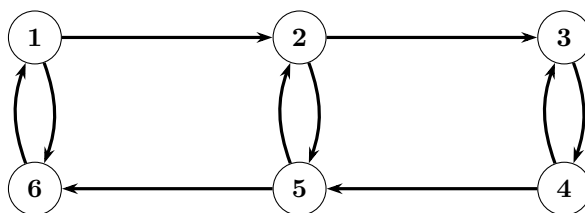


Figure 6: Graph G on two vertices

Example. Consider now $X(G, S^1)$ and $X(G, S^2)$ for $S^1 = \{\{2, 3\}, \{4\}\}$ and $S^2 = \{\{3, 4\}, \{3\}\}$. We claim that these are likely conjugate. What we have done is essentially 'bumped' an element from S_2 to S_1 . Note that in doing this we have preserved word length for all our cycles (in this case we only have one). Simply bumping between the sets of two vertices like this can often lead to conjugacy (as long as we have the lengths of corresponding words preserved properly as stated). However, it cannot be done it just any way. Some cycles interact differently with other cycles in more complex ways, making preserving this word length harder.

In [3] Mattson and Sattler discuss some conditions for bumping elements between sets such as these for $\mathcal{S}$ -limited shifts, which turned out to be very helpful for determining conjugacy. With this in mind, ideas of how we may be able to swap or bump between sets to make conjugate $\mathcal{S}$ -graph shifts may be helpful to consider. We suspect that our non-canceled non-singleton disjoint cycle sets may tell us about how we can swap/bump between our sets and still maintain conjugacy. Consider again the graph posed earlier,



Recall that the non-singleton disjoint cycle set $\{16, 34\}$ was not canceled in our entropy formula. We suspect that this gives us insight into how we may bump between or swap sets on this graph. Say we are considering the shift $X(G, S^1)$ where G is the graph above and $S^1 = \{S_1, S_2, S_3, S_4, S_5, S_6\}$. We conjecture that the set $\{16, 34\}$ tells us that we can freely 'bump' (where bumping is done as shown in our previous example, adding/subtracting elementwise across each set) between S_1 and S_6 , and S_3 and S_4 respectively (yet not S_2 and S_5) while maintaining conjugacy. We made some attempts on this example to determine whether or not this fact was true, obtaining some hopeful evidence that it may be. Entropy calculations and other intuitive checks held. Additionally, this would maintain word length for each cycle. Note that words from the cycle 16 and 34 would certainly maintain the same length. And so would words from the 6 cycle, and our two 4 cycles. So this is hopeful evidence that our claim may have grounds. However, we could not show this rigorously.

The claim that you can *not* bump between 2 and 5 seems to hold even more weight. With some investigation of the graph, allowable transitions, and some consideration of what the conjugacy (sliding block code) would actually look like, it appears that a bump like this would break conjugacy. We know cycles must map to cycles, and everything in between should hopefully get covered by this. The issue arises in the fact that the cycle 25 can actually be 'entered/exited' in two different ways, where as 16 and 34 can only be entered and exited in one way for each. This leads to several potential words that could pose an issue when building our code, and any bumping between 2 and 5 seems to present some impossibilities when considering how our map would look. For example, consider some word where we pass from 2 to 3 and then stay in the 34 cycle for the rest of the word. It is unclear how this would get mapped to when we do some sort of bumping between 2 and 5.

Overall this idea seems to be promising for further consideration. Briefly, we also conjectured that this set may tell us we can do a full swap of the sets between 16 and 34, that is

$$\{S_1, S_2, S_3, S_4, S_5, S_6\} \rightarrow \{S_3, S_2, S_1, S_6, S_5, S_4\}$$

We did not investigate much further into this idea, but it is seemingly an additional thing to consider for the interested researcher.

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